

Francisco M. Garcia

Senior Machine Learning Engineer, Quince

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Education

Doctor of Philosophy

2015-2019

College of Information and Computer Sciences, University of Massachusetts Amherst

Advisor: Philip S. Thomas

Master of Science

2013-2015

College of Information and Computer Sciences, University of Massachusetts Amherst

Advisor: Sridhar Mahadevan

Bachelor of Science

2008-2011

Misher College of Arts and Sciences, University of the Sciences in Philadelphia

Experience

Quince

2024-Present

Senior Machine Learning Engineer

Responsible for building and scaling the recommendations systems running in the store.

Responsible for defining roadmaps around personalization initiatives in the company.

Roku

2022-2024

Senior Machine Learning Scientist

Research in machine learning for recommendations and ranking. Build and deploy models to be used in production.

Amazon Alexa

2019-2022

Applied Scientist – L5

Conduct research and implement solutions for the core natural language understanding system in Alexa.

University of Massachusetts Amherst

2015-2019

Doctoral Student

Conducted research in reinforcement learning with a focus on temporal abstraction and meta-learning.

Microsoft Research

2016

Research Intern

Conducted research in navigation of unmanned air vehicles (UAVs).

Research Intern

Conducted research in natural language techniques to support business analytics teams.

Publications

[NAACL 2021] F. G. Garcia, L. Chen, V. Kumar, H. Xie, J. Lu. Industry Scale Semi-Supervised learning for Natural Language Understanding. In *Proceedings of the 2021 Annual Conference of the North American Chapter of the Association for Computational Linguistics*.

[Ph.D. Dissertation 2020] F. G. Garcia. Improving Reinforcement Learning Techniques by Leveraging Prior Experience.

[NIPS 2019] F. M. Garcia, P. S. Thomas. A Meta-MDP Approach to Exploration for Lifelong Reinforcement Learning. In *Proceedings of the Thirty Third Conference on Neural Information Processing Systems*, December 2019.

[RLDM 2019] F. M. Garcia, C. Nota, P. S. Thomas. Learning Temporal Abstractions from Demonstration: A Probabilistic Approach to Offline Option Discovery. In *The Fourth Multidisciplinary Conference on Reinforcement Learning and Decision Making*, July 2019.

[AAMAS 2019] F. M. Garcia, B. C. da Silva, P. S. Thomas. A Compression-Inspired Framework for Macro Discovery. In *Proceedings of the Eighteenth International Conference on Autonomous Agents and Multiagent Systems*, May 2019. Extended Abstract

[CAVW 2014] K. Ninomiya, M. Kapadia, A. Shoulson, **F. M. Garcia**, N. I. Badler. Planning Approaches to Constraint-Aware Navigation in Dynamic Environments. In *Computer Animation and Virtual Worlds*, 2014.

[ICRA 2014] F. M. Garcia, M. Kapadia, N. I. Badler. GPU-Based Dynamic Search on Adaptive Resolution Grids. In *Proceedings of International Conference on Robotics and Automation*, June 2014

[MIG 2013] M. Kapadia, K. Ninomiya, A. Shoulson, **F. M. Garcia**, N. I. Badler. Constraint Aware Navigation in Dynamic Environments. In *Proceedings of the Sixth International Conference on Motion in Games*, 2013

[IROS 2013] M. Kapadia, **F. M. Garcia**, N. I. Badler. Dynamic Search on the GPU. In *Proceedings of the International Conference on Intelligent Robots and Systems*, 2013

[SCA 2013] M. Kapadia, A. Porres, **F. M. Garcia**, V. Reddy, N. Pelechano, N. I. Badler. Multi-Domain Real-Time Planning in Dynamic Environments. In *EUROGRAPHICS Symposium of Computer Animation*, 2013

[MIG 2011] A. Shoulson, **F. M. Garcia**, M. Jones, R. Mead, N. I. Badler. Parameterizing Behavior Trees. In *Proceedings of the Fourth International Conference on Motion in Games*, 2011

Workshops and Non-peer reviewed Publications

[Arxiv 2020] **F. M. Garcia**, C. Nota, P. S. Thomas. Learning Reusable Options for Multi-Task Reinforcement Learning. <https://arxiv.org/abs/2001.01577>

[AAAI 2019] **F. M. Garcia**, B. C. da Silva, P. S. Thomas. A Compression-Inspired Framework for Macro Discovery. In *Workshop on Reinforcement Learning in Games*, 2019.
Extended version: <https://arxiv.org/abs/1711.09048>

[AAAI 2019] **F. M. Garcia**, P. S. Thomas. A Meta-MDP Approach to Exploration for Lifelong Reinforcement Learning. In *Workshop on Reinforcement Learning in Games*, 2019.
Extended version: <https://arxiv.org/abs/1902.00843>

[Arxiv 2016] S. Giguere, **F. M. Garcia**, S. Mahadevan. A Manifold Approach to Learning Mutually Orthogonal Subspaces. <https://arxiv.org/abs/1703.02992>

Academic Service

I served as reviewer for:

- Association for the Advancement of Artificial Intelligence (AAAI): [2020-2024]
- International Joint Conference in Artificial Intelligence (IJCAI): [2023-2025]
- International Conference on Machine Learning (ICML): [2019-2024]
- Advances in Neural Information Processing Systems (NIPS): [2020-2024]
- International Conference on Autonomous Agents and Multi-agent Systems (AAMAS): [2021]
- International Conference on Learning Representations (ICLR): [2021-2025]
- International Conference on Motions in Games (MIG): 2014
- International Conference on Intelligent Robots and Systems (IROS): 2015

References

Available upon request.